



Evaluating the quality and readability of ChatGPT-generated patient-facing medical information in rhinology

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Abstract

Purpose The artificial intelligence (AI) chatbot ChatGPT has become a major tool for generating responses in healthcare. This study assessed ChatGPT's ability to generate French preoperative patient-facing medical information (PFI) in rhinology at a comparable level to material provided by an academic source, the French Society of Otorhinolaryngology (*Société Française d'Otorhinolaryngologie et Chirurgie Cervico-Faciale*, SFORL).

Methods ChatGPT and SFORL French preoperative PFI in rhinology were compared by analyzing responses to 16 questions regarding common rhinology procedures: ethmoidectomy, sphenoidotomy, septoplasty, and endonasal dacryocystorhinostomy. Twenty rhinologists assessed the clarity, comprehensiveness, accuracy, and overall quality of the information, while 24 nonmedical individuals analyzed the clarity and overall quality. Six readability formulas were used to compare readability scores.

Results Among rhinologists, no significant difference was found between ChatGPT and SFORL regarding clarity (7.61 ± 0.36 vs. 7.53 ± 0.28 ; $p = 0.485$), comprehensiveness (7.32 ± 0.77 vs. 7.58 ± 0.50 ; $p = 0.872$), and accuracy (inaccuracies: 60% vs. 40%; $p = 0.228$), respectively. Non-medical individuals scored the clarity of ChatGPT significantly higher than that of the SFORL (8.16 ± 1.16 vs. 6.32 ± 1.33 ; $p < 0.0001$). The non-medical individuals chose ChatGPT as the most informative source significantly more often than rhinologists (62.8% vs. 39.7%, $p < 0.001$).

Conclusion ChatGPT-generated French preoperative PFI in rhinology was comparable to SFORL-provided PFI regarding clarity, comprehensiveness, accuracy, readability, and overall quality. This study highlights ChatGPT's potential to increase accessibility to high quality PFI and suggests its use by physicians as a complement to academic resources written by learned societies such as the SFORL.

Keywords Artificial intelligence · Accuracy · Readability · Rhinology · Preoperative information

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Introduction

ChatGPT (OpenAI, San Francisco, California, United States of America) remains the most widely used artificial intelligence (AI) chatbot since its release in 2020 [1]. This large language model can generate human-like responses by analyzing and processing extensive amounts of data through techniques such as reinforcement learning from human feedback [2]. In the healthcare domain, AI tools have proven effective in various applications, such as translating medical terminology, answering common surgical questions, and enhancing the readability of medical materials [3–11]. These applications highlight ChatGPT's potential to improve the efficiency of medical consultation and quality of patient-facing medical information (PFI). ChatGPT provides direct answers that are often well-written, detailed, and in if-then format, which give patients access to immediate information while waiting to reach a clinician [12]. This capacity could be helpful especially when alternative resources are limited and may help to reduce well-documented health disparities which exist across the socioeconomic gradient [13–15]. Also, it is important to manage and monitor the ongoing information exchange to ensure that individuals obtain information from reliable sources [16]. In otorhinolaryngology (ORL), where communication between patients and surgeons is critical, high-quality PFI is essential to reduce postoperative morbidity. Indeed, lower health literacy and limited understanding of postoperative instructions have been associated with worse outcomes [17, 18]. This is particularly important for endoscopic sinus surgeries (ESS) and nasal surgeries, which represent approximately 11,500 surgeries per year in France [19], 9,000 in England and Wales [20], and 257,000 in the United States of America [21]. Although these procedures are common, there is often a gap between the readability of PFI created by academic sources, i.e. surgeons who are part of learned societies, and the level required for sufficient patient comprehension. For example, a study by Armache et al. found that the readability of PFI for head and neck cancer exceeded the sixth-grade readability level recommended by the American Medical Association (AMA) [22]. Also, studies consistently show that patient satisfaction and comprehension of current educational materials are suboptimal, with many patients reporting difficulty understanding medical terminology and a lack of clarity in the provided information. Indeed, readability of online information often exceeds the average reading level of the population, limiting accessibility and comprehension [23].

While ChatGPT has demonstrated the ability to generate high-quality PFI in English in ORL-related topics such as septal deviation, acute rhinosinusitis, or epistaxis, only a few studies have evaluated the effectiveness and quality

of ChatGPT-generated French PFI [24–26] and none concerned ChatGPT-generated preoperative PFI in rhinology. Moreover, there are specific challenges in technical medical terminology that need to be overcome through high readability and yet sufficient accuracy since patients who obtained accurate and appropriate information can adapt to their treatment and participate in decision-making processes more easily [27]. More specifically, there is a need to evaluate ChatGPT-generated PFI for ESS and nasal surgeries such as ethmoidectomy, sphenoidotomy, septoplasty, and endonasal dacryocystorhinostomy. This gap in research raises questions about the clarity, accuracy, comprehensiveness and overall quality of ChatGPT-generated PFI in languages other than English, as well as its potential to meet the standards set by learned societies such as the French Society of ORL and Head & Neck Surgery (*Société Française d'ORL et de Chirurgie Cervico-Faciale, SFORL&CCF*), referred to as SFORL in the rest of the manuscript. Addressing these gaps is crucial for determining ChatGPT's ability to produce reliable and understandable PFI across different medical fields.

This study main aim was to evaluate ChatGPT's ability to generate French preoperative PFI in rhinology by comparing ChatGPT-generated PFI with that provided by an academic source, the SFORL. The primary outcome focused on multiple parameters, including clarity, comprehensiveness, accuracy, readability, and the overall quality of the preoperative PFI, both for those generated by ChatGPT and those provided by the SFORL. The secondary outcome aimed to determine whether prior familiarity with the SFORL-provided PFI influenced the evaluation of the quality of PFI generated by ChatGPT. Our main hypothesis was that ChatGPT could generate high-quality French preoperative PFIs in rhinology, encompassing all relevant details, including preoperative information, surgical steps, and key aspects of follow-up care. To address this question, various common rhinology procedures were selected based on their frequency and level of technical complexity, ranging from simpler procedures (e.g., septoplasty) to more intricate ones (e.g., endonasal dacryocystorhinostomy) and more extensive interventions (e.g., ethmoidectomy and sphenoidotomy).

Methods

Sources of patient-facing medical information

The French preoperative PFI provided by the SFORL and validated by the associated academic institution, the French Otorhinolaryngology College (*Collège français d'ORL et de chirurgie cervico faciale, CFORL*), was retrieved. This

PFI, developed by experts in ORL and head and neck surgery, benefits patients scheduled for surgery and internet users seeking information online. Given its extensive distribution among French patients and the legal requirement to include it in the patients' preoperative medical files, the French preoperative PFI provided by the SFORL was compared to ChatGPT-generated French preoperative PFI. The responses produced by ChatGPT and SFORL for four common rhinology procedures—ethmoidectomy, sphenoidotomy, septoplasty, and endonasal dacryocystorhinostomy—were evaluated.

Generating ChatGPT patient-facing medical information

ChatGPT (v.4.0, Open AI, San Francisco, California, United States of America) was utilized due to its improved accuracy and precision in the healthcare setting compared to ChatGPT v.3.5 [28, 29]. In a new conversation, ChatGPT responded to 16 questions extracted from the SFORL-provided PFI for four common rhinology procedures (Supplemental Table 1). The titles and subtitles from each section provided by the SFORL were adapted to form questions, which were then input into a new conversation with ChatGPT. For example, in the ethmoidectomy PFI datasheet, the subtitle goal of the surgery (*but de l'intervention*) was adapted to “What is the goal of an ethmoidectomy? (*quel est le but d'une ethmoïdectomie?*) » to help prompt ChatGPT. To ensure ChatGPT responses were comparable to the information provided by the SFORL, protocols were established to maintain similar word counts and language. The approximate word count of the SFORL information was indicated in the prompt input for ChatGPT. For instance, in a prompt for sphenoidotomy, the following instruction was given: [*Answer the question in French and in 70 words*]. Additionally, to provide understandable preoperative PFI, ChatGPT was asked to write in layman's terms for all prompts: [*Explain the answer to patients in layman's terms*]. Therefore, to effectively generate French preoperative PFI in rhinology, the following prompt was used in a new chat with ChatGPT: [*Can you answer the following question as an education purpose for patients before a surgical procedure: “question?” Answer in French and in # words. Explain in layman's terms*]. There was no medical assessment of ChatGPT responses to assess the quality of the prompt without bias.

Evaluation of responses by rhinologists

French rhinologists, defined as senior ORL surgeons subspecialized in rhinology with experience in diagnosing, treating, and managing patients undergoing ethmoidectomy, sphenoidotomy, septoplasty, and endonasal

dacryocystorhinostomy, blindly evaluated the responses generated by ChatGPT and those provided by the SFORL through an anonymous Google Form. They were all recruited on a voluntary basis among the members of the French Association of Rhinology, a learned society of surgeons engaged in both research and clinical practice in rhinology. The age, sex, and highest level of education of the participating rhinologists were collected. Rhinologists were asked if they were familiar with the SFORL's preoperative PFI in rhinology. Then they were given the 16 questions paired with the responses from both ChatGPT and the SFORL to review but were blinded to the source of the responses, which were randomized. Rhinologists were asked to review the responses provided by ChatGPT and the SFORL regarding clarity and comprehensiveness on a Likert scale of one to ten (1 = *Poor*, 10 = *Excellent*). Additionally, rhinologists marked any responses they deemed inaccurate and were asked to explain their reasoning. Terms were defined to participants as follows: clarity relates to the intelligibility and coherence of the response, comprehensiveness pertains to the breadth and quality of the response, and accuracy refers to the factual correctness of the response. Since multiple inaccuracies could be described by one participant in the same answer, each inaccuracy cited was treated individually and added to the total number reported. Finally, rhinologists determined, for each of the 16 questions, which source they believed was providing the highest quality response. No time limit was enforced to complete the Google Form.

Evaluation of responses by non-medical individuals

French non-medical individuals, defined as individuals without any medical background, blindly evaluated the same set of responses generated by ChatGPT and those provided by the SFORL through an anonymous Google Form. Their age, sex, and highest level of education were collected. Then they were given the 16 questions paired with the responses from both ChatGPT and the SFORL to review but were blinded to the source of the responses, which were randomized. Non-medical individuals were asked to assess the clarity of each response using a Likert scale from 0 to 10 (0 = *Poor*; 10 = *Excellent*). They were also asked to choose, for each of the 16 questions, which source they believed was providing the highest quality response. No time limit was enforced to complete the Google Form.

Readability analysis

Readability scores of ChatGPT-generated and SFORL-provided preoperative PFI in rhinology (16 answers*2 sources) were computed and compared using six validated readability formulas: the Flesch-Kincaid Grade Level (FKGL), Flesch

Reading Ease (FRE), Gunning Fog Index (GFI), Simple Measure of Gobbledygook (SMOG) Index, Coleman-Liau Index (CLI), and Automated Readability Index (ARI) [30–34] (Supplemental Table 2).

Statistical analysis

Continuous variables were expressed as means with standard deviations (SD) and discrete outcomes as absolute and relative (%) frequencies. To assess the quality of ChatGPT-generated French preoperative PFI in rhinology, ChatGPT and the SFORL preoperative PFI were compared regarding clarity (according to both rhinologists and non-medical individuals), comprehensiveness and accuracy (according to rhinologists only), readability scores (based on six validated scales), and overall quality (according to rhinologists and non-medical individuals) by using either paired t-tests or Fisher's exact tests for continuous or discrete data, respectively. A subgroup analysis was also planned to assess for potential bias within the rhinologist group, between those familiar with the SFORL PFI and those who were not. Also, inter-rater reliability was assessed for rhinologists.

Regarding accuracy comparison, a verbatim analysis was performed based on the comments made in the free text. To that end, six subgroups of inaccuracies were planned as follows: true inaccuracies, insufficient clarity, insufficient comprehensiveness, translation language issues, general remarks, and comments reporting differences from common practice. The verbatim analysis of all inaccuracies reported was performed based on the total number of potential answers from all the rhinologists ($20 \times 16 = 320$). Significance was defined as $p < 0.05$. The statistical analyses were performed using the R software (v. 4.3.3, R Foundation for Statistical Computing, Vienna, Austria, www.r-project.org).

Table 1 Demographics of rhinologists and non-medical individuals

Characteristics	Rhinologists (<i>n</i> = 20)	Non-medical Individuals (<i>n</i> = 24)
Male, % (n)	70 (14/20)	50 (12/24)
Age, years, mean (SD)	32.55 (± 3.61)	37.46 (± 10.48)
Highest level of education		
High school graduate, % (n)	0.00 (0/20)	4.15 (1/24)
College graduate, % (n)	0.00 (0/20)	91.70 (22/24)
Trade school, % (n)	0.00 (0/20)	4.15 (1/24)
Medical school, % (n)	20.00 (20/20)	0.00 (0/24)

Footnotes Continuous variables are expressed as mean with standard deviations (SD) and discrete outcomes as absolute and relative (%) frequencies

Results

Population included

Twenty French rhinologists and 24 French non-medical individuals answered a questionnaire. All participants completed the entire Google form and there were no missing data. The age, sex, and level of education of both rhinologists and non-medical individuals are reported in Table 1.

Clarity

There was no significant difference regarding clarity between PFI generated by ChatGPT and that provided by the SFORL according to rhinologists (7.61 ± 0.36 vs. 7.53 ± 0.28 , $p = 0.485$; Fig. 1A). The subgroup analysis according to whether rhinologists were familiar or not with the SFORL PFI showed no significant difference (7.33 ± 0.33 vs. 7.28 ± 0.26 ; $p = 0.81$, and 7.91 ± 0.43 vs. 7.81 ± 0.31 , $p = 0.808$, respectively; Fig. 1A). There was a significant difference in clarity between ChatGPT and SFORL PFI according to non-medical individuals who favored ChatGPT-generated preoperative PFI in rhinology (8.16 ± 1.16 vs. 6.32 ± 1.33 , $p < 0.0001$; Fig. 1A).

Comprehensiveness

Regarding comprehensiveness, there was no significant difference between the preoperative PFI from ChatGPT and that of the SFORL according to rhinologists (7.32 ± 0.77 vs. 7.58 ± 0.50 , $p = 0.872$; Fig. 1B). The subgroup analysis according to whether rhinologists were familiar or not with the SFORL PFI showed no significant difference (7.11 ± 0.69 vs. 7.42 ± 0.54 , $p = 0.748$; and 7.57 ± 0.88 vs. 7.86 ± 0.59 , $p = 0.52$, respectively; Fig. 1B).

Accuracy

In terms of accuracy, 60% (12/20) of rhinologists found at least one inaccuracy in the PFI generated by ChatGPT vs. 40% (8/20) in the PFI provided by the SFORL ($p = 0.228$). Among the 320 answers from rhinologists, more inaccuracies were noted for ChatGPT than the SFORL (37.8%, 121/320 vs. 16.8%, 54/320, $p < 0.001$). The verbatim analysis found that for ChatGPT-generated PFI, rhinologists reported true inaccuracies (48/121), insufficient clarity (26/121), insufficient comprehensiveness (14/121), translation language problems (11/121), general remarks (14/121), and information that differed from common practice (8/121). For the SFORL-provided PFI, rhinologists reported true inaccuracies (4/54), insufficient clarity (12/54), insufficient comprehensiveness (10/54), translation language problems

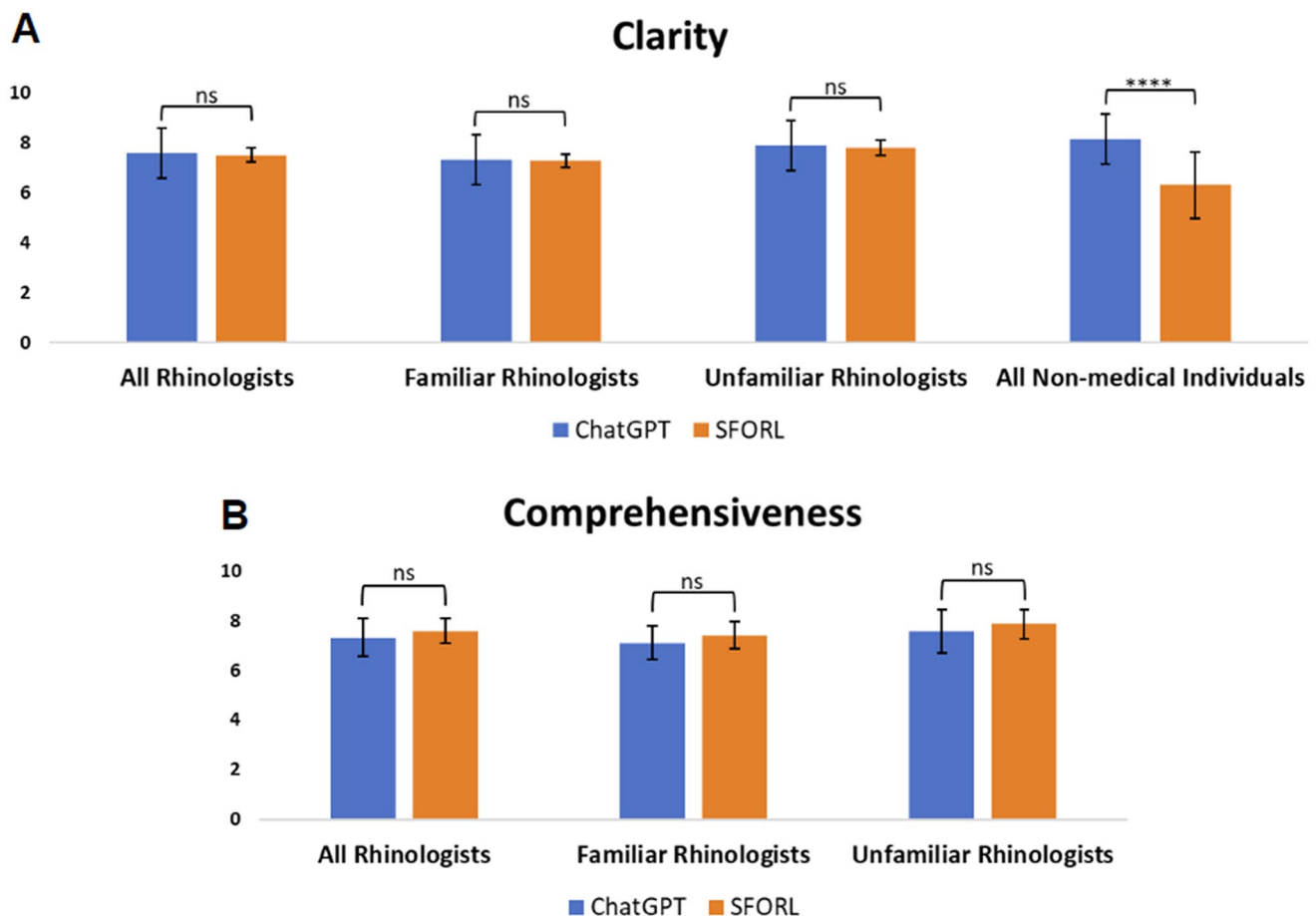


Fig. 1 Rhinologists and non-medical individuals' evaluations of clarity and comprehensiveness of ChatGPT and SFORL patient-facing information in rhinology. Abbreviations: ns: no significance; ****: $p < 0.0001$

(0/54), general remarks (14/54), and information that differed from common practice (14/54). Details are in Table 2 and supplemental Table 1 (Supplemental Table 1).

Regarding true inaccuracies, there was a significant difference in favor of SFORL-provided PFI compared to ChatGPT-generated PFI with 98.7% (316/320) of correct responses vs. 85.0% (272/320, $p < 0.001$, respectively). Among the inaccuracies reported, because several authors reported the same ones, there were 6 different true inaccuracies raised by rhinologists in the ChatGPT-generated PFI (corresponding to 98.1% of correct responses) compared to 3 in the SFORL-provided PFI (corresponding to 99.1% of correct responses, $p = 0.504$). The 6 true inaccuracies reported by rhinologists in ChatGPT-generated PFI, were “*améliorer la respiration*” (improve breathing in sphenoidotomy and ethmoidectomy) in question 1 (Q1) and Q5, “*endoscope flexible*” (use of a flexible endoscope to perform any sinus surgery) in Q2, “*fuite de liquid céphalorachidien à long terme*” (cerebrospinal fluid leak as a long-term complication) in Q3 and Q7, and “*accumulation de sang sous la peau*” (skin hematoma in septoplasty) in Q11. The 3 true

inaccuracies reported by rhinologists in the SFORL-provided PFI were “*infection des oreilles*” (ear infection after septoplasty) in Q9 by two rhinologists, “*larmoiements lié à l'irritation des voies lacrymales*” (epiphora after septoplasty by lacrimal duct breach) in Q11, and “*cède généralement après un geste de mouchage*” (epistaxis that stopped spontaneously or just by blowing your nose after dacryocystorhinostomy) in Q15.

Readability

There was no significant difference in terms of readability between the ChatGPT and SFORL PFI, no matter which readability test formula was implemented (Table 3). All the results exceeded the recommended readability level of 6th to 8th grade.

Overall quality

Among the 384 potential answers from non-medical individuals (24 non-medical individuals*16 answers), 62.8%

Table 2 ChatGPT and SFORL inaccuracy results

Topic/ question	Inaccuracies of ChatGPT		Inaccuracies of SFORL	
	Rhinologists % (n)	Classification	Rhinologists % (n)	Classification
Ethmoid Q1	80% (16/20)	Insufficient comprehensiveness: 3 Translation issues: 6 True inaccuracy: 7	25% (5/20)	Insufficient comprehensiveness: 1 Insufficient clarity: 2 General remarks: 2
Ethmoid Q2	80% (16/20)	Insufficient comprehensiveness: 1 Translation issues: 1 General remarks: 2 True inaccuracy: 15	40% (9/20)	Insufficient clarity: 4 Differ from common practice: 5
Ethmoid Q3	70% (14/20)	Insufficient comprehensiveness: 1 Insufficient clarity: 7 General remarks: 1 True inaccuracy: 7	40% (8/20)	Insufficient comprehensiveness: 5 Insufficient clarity: 2 Differ from common practice: 1
Ethmoid Q4	5% (1/20)	General remarks: 1	15% (3/20)	Insufficient comprehensiveness: 1 General remarks: 2
Sphenoid Q5	65% (13/20)	Insufficient comprehensiveness: 1 General remarks: 1 True inaccuracy: 11	10% (2/20)	Insufficient clarity: 1 General remarks: 1
Sphenoid Q6	50% (10/20)	Insufficient clarity: 2 Translation issues: 2 Differ from common practice: 8	15% (3/20)	Differ from common practice: 3
Sphenoid Q7	35% (7/20)	Insufficient comprehensiveness: 2 Insufficient clarity: 1 General remark: 1 True inaccuracy: 4	20% (5/20)	Insufficient comprehensiveness: 2 Insufficient clarity: 1 General remarks: 2
Sphenoid Q8	15% (3/20)	Insufficient comprehensiveness: 3	0% (0/20)	/
Septoplasty Q9	20% (4/20)	Insufficient clarity: 1 Translation issues: 2 General remarks: 1	20% (4/20)	General remarks: 3 True inaccuracy: 2
Septoplasty Q10	25% (5/20)	Insufficient clarity: 3 General remarks: 2	25% (5/20)	Differ from common practice: 5
Septoplasty Q11	30% (6/20)	Insufficient comprehensiveness: 1 Insufficient clarity: 2 True inaccuracy: 4	5% (1/20)	General remarks: 1 True inaccuracy: 1
Septoplasty Q12	40% (8/20)	Insufficient comprehensiveness: 2 Insufficient clarity: 7	10% (2/20)	Insufficient clarity: 1 General remarks: 1
ED Q13	15% (3/20)	General remarks: 3	0% (0/20)	/
ED Q14	10% (2/20)	General remarks: 2	0% (0/20)	/
ED Q15	0% (0/20)	/	25% (5/20)	Insufficient comprehensiveness: 1 Insufficient clarity: 1 General remarks: 2 True inaccuracy: 1
ED Q16	15% (3/20)	Insufficient clarity: 3	0% (0/20)	/

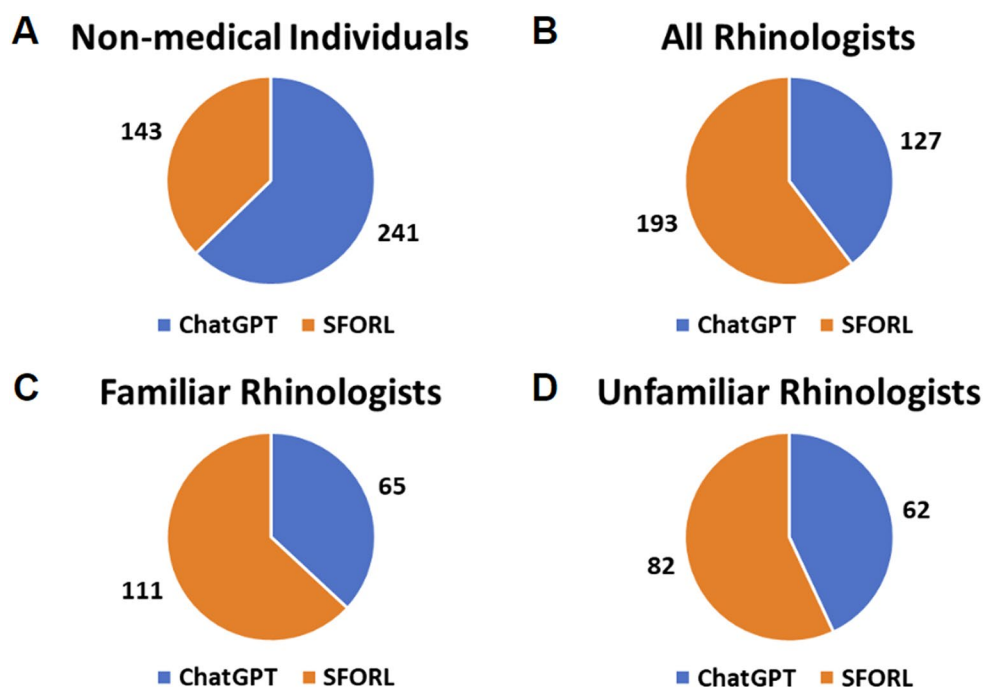
Abbreviations ED: Endonasal dacryocystorhinostomy, Q: Question

Table 3 ChatGPT and SFORL readability results

Index	ChatGPT mean (Standard deviation)	ChatGPT level of education	SFORL mean (Standard Deviation)	SFORL level of education
FRE	31.69 (± 5.03)	College	28.55 (± 4.97)	College graduate
FKGL	12.90 (± 0.91)	12th grade	13.21 (± 0.59)	College freshman
GFI	14.68 (± 0.70)	College sophomore	15.29 (± 0.60)	College junior
SMOG	14.77 (± 0.58)	College sophomore	15.10 (± 0.42)	College sophomore
CLI	13.89 (± 1.49)	College freshman	14.38 (± 1.56)	College sophomore
ARI	11.99 (± 1.61)	10th grade	12.18 (± 0.82)	11th grade

Footnotes Readability results are given with the mean and standard deviations for each scale for ChatGPT-generated and SFORL-provided PFI as well as the corresponding level of educations for both ChatGPT and SFORL. Abbreviations FRE, Flesch Reading Ease; FKGL, Flesch-Kincaid Grade Level; GFI, Gunning Fog Index; SMOG, Simple Measure of Gobbledygook; CLI, Coleman-Liau Index; and ARI, Automated Readability Index

Fig. 2 Preference of non-medical individuals (A) and rhinologists (B) familiar (C) or unfamiliar (D) with SFORL information for patient-facing information in rhinology generated by either ChatGPT or SFORL. Footnotes: Number of non-medical individuals (A), overall rhinologists (B), rhinologists familiar (C) and unfamiliar (D) with the SFORL PFI preferring ChatGPT-generated PFI or SFORL-provided PFI



(241/384) chose ChatGPT as the most informative source compared to SFORL (Fig. 2A). Among the 320 potential answers from rhinologists (20 rhinologists*16 answers), 39.7% (127/320) selected ChatGPT as the highest-quality source of preoperative PFI in rhinology (Fig. 2B). The non-medical individuals chose ChatGPT as the most informative source significantly more often than rhinologists ($p < 0.001$). Among familiar rhinologists (11*16 potential answers), 36.9% (65/176) chose ChatGPT, and among unfamiliar rhinologists (9*16 potential answers), 43.1% (62/144) chose ChatGPT as the highest-quality source of preoperative PFI in rhinology (Fig. 2C-D) with no significant difference between familiar and unfamiliar rhinologists ($p = 0.301$). Inter-rater reliability was assessed for all answers from the 20 rhinologists and the kappa coefficient was 0.004 ($n = 20$, answers = 64, $p = 0.419$).

Discussion

ChatGPT can generate high-quality French preoperative PFI in rhinology. There was no significant difference in clarity, comprehensiveness, accuracy, readability, and overall quality of the preoperative PFI generated by ChatGPT compared to the one provided by the SFORL. There was also no significant difference between rhinologists familiar or unfamiliar with the SFORL preoperative PFI. Conversely, nonmedical individuals scored the clarity of ChatGPT-generated PFI in rhinology higher than that of the SFORL-provided PFI.

In terms of clarity, ChatGPT-generated French preoperative PFI was comparable to that of the SFORL according

to rhinologists but better according to non-medical individuals, illustrating the user-friendliness of ChatGPT-generated PFI compared to those provided by learned societies. Recent studies also reported a higher clarity of Chat-GPT, which conversely to the results herein, was based on the evaluation by plastic and reconstructive surgeons (PRS). Berry et al. reported that the clarity scores of ChatGPT-generated PFI in microsurgery were significantly higher than the ones generated by the American Society of Reconstructive Microsurgery based on answers coming from 6 PRS [4]. In addition, Fazilat et al., reported higher clarity scores for ChatGPT-generated PFI in cleft lip and palate surgery according to 11 PRS, when compared to PFI from academic/professional sources [29]. Regarding comprehensiveness, the French preoperative PFI generated by ChatGPT was comparable to the SFORL-provided preoperative PFI as assessed by rhinologists. These findings align with the results from Xie et al. who found AI-generated PFI for rhinoplasty to be comprehensive ($n = 4$ PRS) [35] and those of Seth et al. who reported that ChatGPT produces coherent and comprehensive PFI for breast augmentation ($n = 3$ PRS) [2]. These results are also supported by the study of Jo et al. ($n = 10$ respondents, 5 healthcare professionals and 5 patients), who found that ChatGPT v4-generated colorectal cancer surgery PFI was comparable to answers provided by a colorectal cancer information book [36].

While ChatGPT was found to produce clear PFI, it was also found to generate more total inaccuracies than the SFORL, even if this difference was not statistically significant. 60% of rhinologists encountered at least one inaccuracy in the ChatGPT-generated preoperative PFI, in comparison

to 40% in the SFORL-provided PFI. However, upon reviewing each inaccuracy reported by rhinologists for ChatGPT, the majority did not qualify as true inaccuracies. Rhinologists noted issues such as insufficient clarity, translation language problems, or simply highlighted information that differed from common practice. For example, the ChatGPT PFI mentioned a short hospitalization after sphenoidotomy instead of outpatient surgery, which represents the actual common practice in some hospitals in France. Additionally, the time of surgery was mentioned as inaccurate for some rhinologists when provided by ChatGPT, but this parameter is rhinologist dependent. Moreover, the 48 true inaccuracies reported (corresponding to 85% of correct responses) only represented six different true inaccuracies (corresponding to 98% of correct responses). These findings, although slightly better, align with the results of Chaker et al. ($n=2$ PRS), who reported that 69% of ChatGPT-generated responses to cleft lip and palate postoperative questions were accurate [37]. Additionally, a study by Gomez-Cabello et al. ($n=3$) also found that ChatGPT-generated postoperative PFI provided high levels of accuracy based on the Likert scale [38]; nevertheless, these two studies were based on a small number of participants. In light of these findings, which highlight that both ChatGPT and the SFORL still produce a few true inaccuracies, rhinologists should always review written PFI before sharing it with patients to provide high quality personalized healthcare. This would allow them to adapt the information to align with their standard practices and deliver tailored information suited to each patient's needs.

Regarding readability, both ChatGPT and SFORL did not generate preoperative PFI in rhinology at the 6th to 8th grade reading level recommended by the NIH and AMA [39]. To increase the readability of ChatGPT-generated PFI, the sentence [*Explain the answer to patients in layman's terms*] was added to the prompt inputted. This was done based on the study by Li et al. who had previously found the prompt successful when asking ChatGPT to alter radiology reports for patients' comprehension [40]. Similar results were reported by Wang et al., who utilized the prompt [*Rephrase this article to a 5th-grade readability level: Article*], which resulted in a significantly improved readability of plastic surgery PFI, although without reaching the 5th-grade reading level [39]. However, SFORL-provided PFI also exceeded the recommended reading level grade meaning that some very specific surgical procedures can be hard to explain in details at such a reading level grade. Similarly, Yurdakurban et al. showed that a college-level education is required to comprehend ChatGPT-generated orthognathic surgery PFI, according to the SMOG index [41]. Furthermore, Berry et al. and Fazilat et al. also found that microsurgery, cleft lip, and cleft palate PFI generated by ChatGPT and academic sources both surpassed the 8th-grade reading

level [29, 42, 43]. After comparing the readability results obtained in the present study to those of Li et al. and Wang et al., a possible explanation for the high reading level grade required to understand ChatGPT-generated PFI herein is that ChatGPT struggles to produce readable information when generating PFI from scratch [39, 40]. Instead, if the text is inputted into ChatGPT and prompted to make the material more readable, ChatGPT can sufficiently alter the information, as seen in the studies by Li et al. and Wang et al. [39, 40]. One limitation identified herein is thus the inability of ChatGPT to generate preoperative PFI in rhinology from scratch at a readability level recommended by the NIH and AMA. Nevertheless, since there was no statistically significant difference in the readability of PFI generated by ChatGPT compared to the PFI from the SFORL, ChatGPT remains comparable to SFORL, demonstrating its value in creating high-level quality PFI with equal readability levels of information sheets developed by rhinologists from learned societies. However, both resources must still improve the readability of their PFI to uphold health literacy, allowing patients to understand and engage with medical information, representing the single best predictor of health status [39, 42].

Regarding overall quality, rhinologists and non-medical individuals' preferences for ChatGPT-generated or SFORL-provided PFI differed significantly, rhinologists preferred SFORL-provided PFI where as non-medical individuals favored ChatGPT-generated PFI. Previous studies in microsurgery and craniofacial surgery have demonstrated that surgeons and patients preferred English ChatGPT information over academic sources [4, 29]. However, in the present study, rhinologists highlighted language translation issues or vagueness from French ChatGPT-generated PFI, which could have led them to prefer the SFORL. Rhinologists reported 11 translation issues in ChatGPT responses compared to zero translation issues in SFORL. Therefore, ChatGPT may be less equipped to generate PFI in French than English.

The present study has limitations. First, it was limited to the perspectives of 20 rhinologists and 24 non-medical individuals who spoke French, all coming from France. Additionally, ChatGPT was the only AI tool utilized herein, even though other programs, such as Gemini AI (Google, Mountain View, California, United States of America), exist. Also, most physicians are now familiar with ChatGPT and have used it before, as it has been publicly available for almost two years. Thus the responses generated by ChatGPT are given in a way that is now easy to recognize, and this may have led to bias in grading, as there is no way to be truly blind for this. Furthermore, different responses are generated when an identical question is inputted into multiple ChatGPT conversations. Therefore, further research into

analyzing the accuracy across all possible responses from ChatGPT could verify the reliability of the AI tool. Also, physicians need to take into account that adoption of stand-alone eHealth tools like ChatGPT is low among persons in lower socioeconomic groups and AI tool may not be used by all patients [13–15].

Since ChatGPT is still a novel chatbot, it is not yet able to act as an independent healthcare resource in the absence of physician oversight. However, ChatGPT could be used by physicians as a tool to help create high quality preoperative PFI in rhinology as it generates clear, comprehensive, accurate, and readable PFI comparable to the SFORL-provided PFI. Leveraging ChatGPT 4.0, trained on high-quality, evidence-based sources under physician supervision, offers a powerful tool to enhance both the readability and accuracy of pre-existing PFI and to develop new ones. Physicians should embrace this approach, as increasing documentation and other responsibilities often reduce the time available for patient education and addressing questions [44]. As patients increasingly rely on the internet without the tools to discern reliable information, the synergy of human expertise and AI can bridge this gap, delivering the accurate, tailored content patients need [45]. Bridging the gap between patients literacy and technical medical terminology will improve patient's health status as it is well-known that poor literacy is associated with poor health status [46]. With continuous training on credible data, ChatGPT's potential extends beyond preoperative PFI in rhinology, supporting patients and physicians across a broader spectrum of healthcare needs.

Conclusion

ChatGPT-generated French patient-facing preoperative medical information in rhinology was comparable to SFORL regarding clarity, comprehensiveness, accuracy, readability, and overall quality. Interestingly, non-medical individuals preferred ChatGPT-generated PFI while rhinologists favored SFORL-provided PFI. This study highlights ChatGPT's potential in increasing accessibility to high quality French preoperative PFI in rhinology and suggests its use by physicians as a complement to academic resources written by learned societies.

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